

Multivariate Lab-3

Date: 14-02-2026

You are given the random vector $X'=[X_1, X_2, X_3, X_4]$ with mean vector $\mu'_x=[4,3,2,1]$ and variance-covariance matrix

$$\Sigma_x = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$$

Partition of X as $X = \begin{bmatrix} X_1 \\ X_2 \\ \dots \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} X^{(1)} \\ \dots \\ X^{(2)} \end{bmatrix}$

Let $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ and consider the linear transformation $AX^{(1)}$ and $BX^{(2)}$.

Find (a) $E(X^{(1)})$, (b) $E(AX^{(1)})$ (c) $Cov(X^{(1)})$ (d) $Cov(AX^{(1)})$ (e) $E(X^{(2)})$ (f) $E(BX^{(2)})$ (g) $Cov(X^{(2)})$ (h) $Cov(BX^{(2)})$ (i) $Cov(X^{(1)}, X^{(2)})$ (j) $Cov(AX^{(1)}, BX^{(2)})$.

#mean vector

```
x<-matrix(data = c(4,3,2,1),ncol=1)
```

#covariance matrix

```
sigma<-matrix(data = c(3,0,2,2,
                        0,1,1,0,
                        2,1,9,-2,
                        2,0,-2,4),ncol=4)
```

```
a<-matrix(data = c(1,2),ncol=2)
```

```
b<-matrix(data = c(1,2,
                  -2,-1),nrow=2)
```

Partition of the X as mu1 and mu2

#partition of the x as mu1 and mu2

```
mu1<-x[1:2]
mu2<-x[3:4]
```

Partition of the sigma as s11,s12,s22

```
s11<-sigma[1:2,1:2]
s22<-sigma[3:4,3:4]
s12<-sigma[1:2,3:4]
```

Linear Transformation

#linear transformation

```
ax1<-a*%mu1
bx2<-b*%mu2
```

Task perform according to the given questions:

#task perform according the given question

```
E_x1<-mu1
E_ax1<-ax1
Cov_x1<-s11
Cov_ax1<-a**s11**t(a)
```

```
E_x2<-mu2
E_bx2<-bx2
Cov_x2<-s22
cov_bx2<-b**s22**t(b)
cov_ax1_bx2<-a**s12**t(b)
```

```
E_x1
```

```
## [1] 4 3
```

```
E_ax1
```

```
##      [,1]
## [1,]    10
```

```
Cov_x1
```

```
##      [,1] [,2]
## [1,]    3    0
## [2,]    0    1
```

```
Cov_ax1
```

```
##      [,1]
## [1,]    7
```

```
E_x2
```

```
## [1] 2 1
```

```
E_bx2
```

```
##      [,1]
## [1,]    0
## [2,]    3
```

```
Cov_x2
```

```
##      [,1] [,2]
## [1,]    9   -2
## [2,]   -2    4
```

```
cov_bx2
```

```
##      [,1] [,2]
## [1,]   33   36
## [2,]   36   48
```

```
s12
```

```
##      [,1] [,2]
## [1,]    2    2
## [2,]    1    0
```

```
cov_ax1_bx2
```

```
##      [,1] [,2]
## [1,]    0    6
```